Sprint Work Assignment

AI Requirements Conflict Detector — Mapped to Frontend / Backend / Testing Pods

IT314 — Software Engineering | Built on the 14-week plan (Sprint 0 to Sprint 7), revision 5

1. Team Roster & Standing Ownership

Each member owns one area for the whole project and reviews all code in it. The project is now a collaborative requirements workspace: teams join with a code, write requirements, see conflicts on a live map, and resolve them through issues and pull-request-style change requests, with AI suggestions that people decide on. Every project starts with compulsory context (an initial SRS or a brief) and ends with an updated SRS written by the LLM and approved by people. Testing remains a dedicated function in every sprint, and every developer writes unit tests for their own code.

Pod	Name	Standing Ownership
Backend	Bansil	Ingestion & authoring: extraction, project context and brief, requirement posts and drafts, owners, change-request intake, raised concerns, updated SRS generator and export, open-items report
Backend	Heet	Embeddings & retrieval: pgvector, candidate filter, live (incremental) retrieval, similar-requirement hints, relationship queries, attribute index for rule checks
Backend	Keval	Classification & LLM: prompts and the project context pack, 5-category classifier, confidence and severity, LLM budget, lint catalogue, pre-flight check, suggestion and SRS section prompts, attribute extraction
Backend	Madhav	Clarification & consequences: questions, impact analysis, change-request checks and findings delta, AI resolution suggestions, notifications, join codes
Backend	Parth	Backend lead & architecture: auth, permissions and roles, review state, lifecycle, decision rules, agreement checks and merge, versions, baselines, audit; reviews all migrations
Frontend	Pranshu	Frontend lead: app shell, login, channels, requirement composer, requirements table, issues, dashboard, baselines, Updated SRS screen, decision-rules settings, change log
Frontend	Abhijeet	Members, roles and join-code screens; project setup wizard; findings and review; live map; change request page; AI suggestion panel; inbox; compliance screens
Frontend / DevOps	Dhavni didi	DevOps & real-time: Docker, CI/CD, Celery queues, WebSocket gateway, graph API and live-map updates, issues backend, written change log, deployment
Testing	Abhyudit	Accuracy & backend testing: planted-conflict benchmark (including reworded conflicts), recall reports, suggestion correctness study, API and database tests
Testing	Affan	E2E, performance & security testing: Playwright, permission tests, join-code abuse tests, WebSocket and load tests, accessibility

The Scrum Master rotates each sprint: Parth (S0–S1), Dhavni didi (S2), Bansil (S3), Pranshu (S4), Heet (S5), Keval (S6), Madhav (S7). The team names one fixed Product Owner proxy for the semester.

2. Plan at a Glance

14 weeks: a one-week Sprint 0, six two-week sprints, and a release week. 224 points are scheduled.

Sprint	Weeks	Goal	Points
S0	1	Re-baseline: new database schema, technical spikes, interface contracts, prototypes	0

Sprint	Weeks	Goal	Points
S1	2–3	Join with a code, roles, workspaces and projects with compulsory context, SRS upload and extraction	37
S2	4–5	Detection engine, real-time channels, custom roles	38
S3	6–7	Requirement composer, live analysis, live map, review — **Integration Checkpoint 1**	37
S4	8–9	Issues (AI and human-raised), change requests, findings delta, agreement rules	37
S5	10–11	AI suggestions with structured feedback, decision rules, rule-based checks — **Integration Checkpoint 2**	38
S6	12–13	Written change log, map preview, LLM-generated updated SRS with review and export, open-items report, compliance	37
S7	14	Release: bug bash, demo data, rehearsal, final report, **final demo**	0

3. Sprint 0 — Re-baseline (Week 1)

No user stories delivered this sprint — schema, contracts and risky technology are proven first.

Name	Task
Parth	Migrations for projects, memberships, roles and permissions (9 default roles seeded); permission-check skeleton; existing access and isolation tests passing on it
Dhavni didi	WebSocket + Redis spike; map-library spike with Abhijeet (500 nodes, 2,000 edges); live-map event format; Docker Compose and CI for new modules
Bansil	Composer field schema and drafts model with Pranshu; EARS sentence templates with Keval; project brief schema (required and optional fields); plan for filtering headings and prose out of extraction
Pranshu	App shell, routing and shared component kit; start login and sign-up-with-code screens
Heet	Confirm embedding model; measure similar-requirement hint speed; draft the live-analysis retrieval contract
Keval	LLM client adapter; first prompts with delimited inputs and a project-context slot; lint (vague-word) catalogue v1; LLM budget design
Madhav	Join-code design (format, hashing, limits, approval); impact-report and checks schema
Abhijeet	Turn the interface wireframes into clickable prototypes for user testing; map-library spike with Dhavni didi
Abhyudit	Benchmark plan: planted-conflict SRS, 40 post-style requirements, 10 scripted change requests, and conflicts worded differently enough to slip past the similarity filter
Affan	Playwright set-up in CI; permission test matrix (every permission × role)

4. Sprint 1 — Joining, Roles, Project Context & Ingestion (Weeks 2–3)

Stories: US-19 (Auth, sign-up only with a code), US-27 (Roles & permissions), US-24 (Workspaces & projects), US-25 (Join with a code), US-60 (Project context required), US-01 (Upload), US-02 (Extraction), US-03 (Progress), US-04 (Requirements list) — 37 points

Name	Task
Parth	US-19 JWT auth with sign-up only by code; US-27 permissions and default roles; US-24 workspaces and projects; US-01 upload into a project; US-04 list API (primary owner); no analysis while a project is in Setup
Madhav	US-25 join codes: create, expiry, use limit, optional approval, revoke (primary owner)
Bansil	US-02 extraction and segmentation, keeping printed IDs; US-60 project context: brief, versions, AI-drafted brief from the initial SRS, activation (primary owner)

Name	Task
Dhavni didi	US-03 progress tracking on Celery stages; event-bus base for real-time
Heet	Embedding stage and vector store (groundwork for US-05)
Keval	US-60 context pack added to every prompt (at most 1,500 tokens); classifier skeleton, output schemas and retries
Pranshu	Login and sign-up-with-code UI; upload, progress and requirements-list UI (US-01, US-03, US-04)
Abhijeet	US-24 workspace and project screens; US-60 project setup wizard and Setup banner; US-25 join-code and join-request screens
Abhyudit	Label the benchmark; API tests for upload and extraction
Affan	US-27 permission tests; join-code expiry, use-limit and brute-force tests; E2E create code → join → upload

Exit check: a project manager creates a project with its initial SRS or brief (analysis stays off until confirmed), creates a code, and a developer joins and sees the requirements.

5. Sprint 2 — Detection Engine, Channels & Custom Roles (Weeks 4–5)

Stories: US-05 (Detect issues — 13 pt core story), US-07 (Explain / confidence), US-26 (Remove members, change roles), US-49 (Custom roles), US-28 (Channels), US-29 (Real-time messaging), US-22 (Prototype validation) — 38 points. One of the two heaviest sprints (see Section 11).

Name	Task
Keval	US-05 five-category classification and US-07 confidence and rationale (primary owner)
Heet	US-05 embeddings and candidate-pair pre-filter (primary owner)
Parth	US-26 remove members and change roles; US-49 custom roles; per-pair classification status; analysis orchestration
Dhavni didi	US-29 WebSocket gateway and real-time messaging (primary owner); first graph API for the map
Madhav	US-28 channels backend; clarification-question prompt drafts
Bansil	Requirement-post and drafts backend groundwork; remove headings and prose from extraction
Pranshu	US-28 channel UI; US-29 chat UI with threads and mentions
Abhijeet	US-07 findings UI (side-by-side, confidence); US-26 and US-49 members and roles screens; US-22 prototypes
Abhyudit	Critical: first classification accuracy run (precision and recall) on the planted-conflict SRS; US-22 sessions
Affan	WebSocket tests; custom-role tests; US-22 usability sessions with sample users

US-05 milestones: conflicts by day 4, duplicates and ambiguous by day 7, incomplete and dependent by day 9.

6. Sprint 3 — Composer, Live Analysis & Live Map (Weeks 6–7) — Integration Checkpoint 1

Stories: US-30 (Composer with EARS templates), US-31 (Live analysis), US-14 (Live requirements map), US-37 (Lifecycle), US-36 (Owners & groups), US-09 (Clarification questions), US-10 (Human review) — 37 points

Name	Task
Bansil	US-30 composer backend (templates, drafts, posting); US-36 owners, stakeholder groups and ownership requests (primary owner)

Name	Task
Heet	US-31 live retrieval: compare each new requirement with the agreed baseline (primary owner)
Keval	US-31 classification of posted requirements; LLM budget guard
Dhavni didi	US-31 analysis queue and events; US-14 live-map updates and one-line written feed (primary backend owner)
Parth	US-37 requirement lifecycle states; US-10 review-state API with the protected-requirement guard (primary owner)
Madhav	US-09 clarification questions on findings and result cards (primary owner)
Pranshu	US-30 composer UI; US-36 owner and group UI
Abhijeet	US-14 live map UI (primary owner); US-10 review UI
Abhyudit	Checkpoint 1 accuracy report: precision, recall and F1 per category, document-style and post-style, with and without the context pack
Affan	Checkpoint 1 E2E for both paths; map updates within 2 s; tenant isolation over REST and WebSocket

Checkpoint 1 demo: join with a code → write a requirement → result card within 60 s → every open map shows the new node and its conflict.

7. Sprint 4 — Issues & Change Requests (Weeks 8–9)

Stories: US-11 (Impact engine), US-33 (Change request), US-34 (Checks & findings delta), US-35 (Review & merge), US-52 (Issues & auto conflict issues), US-12 (Version history), US-56 (Raise a concern) — 37 points

Name	Task
Madhav	US-11 impact analysis; US-34 consequence checks and findings delta (resolved / new / still open / must re-run) (primary owner); suggestion schema
Heet	US-11 relationship queries
Keval	US-34 AI summary of trade-offs
Bansil	US-33 change requests with base version and proposal revisions; US-56 human-raised concerns backend (primary owner)
Parth	US-35 review and merge with the agreement checks (stale approvals dismissed) (primary owner); US-12 version history
Dhavni didi	US-52 issues and automatic conflict issues for two owners (primary owner)
Pranshu	US-52 issues UI; US-12 history UI; US-56 Raise-a-concern form and AI / Rule / Person source badges
Abhijeet	US-33 change editor; US-34 Checks and findings-delta tabs; US-35 review and merge screen with agreement checklist
Abhyudit	Check affected requirements and findings delta for each scripted change request; auto-issue owners
Affan	Merge rules: no self-approval, compliance sign-off, needs-update, approvals dismissed when a proposal is edited

8. Sprint 5 — Suggestions, Decision Rules & Rule Checks (Weeks 10–11) — Integration Checkpoint 2

Stories: US-45 (AI suggestions with structured feedback), US-57 (Decision rules), US-58 (Rule-based cross-checks), US-38 (Inbox), US-13 (Dashboard), US-40 (Baseline snapshots), US-51 (Lint & pre-flight), US-06 (Thresholds) — 38 points

Name	Task
Madhav	US-45 suggestion engine with six structured feedback outcomes (primary owner); US-38 notifications backend
Keval	US-45 prompts; US-51 lint, pre-flight check and AI wording; US-58 attribute extraction
Heet	US-58 attribute index and rule checks for numbers, negation, actors and conditions (primary owner); US-06 similarity threshold
Parth	US-57 decision rules engine (primary owner); US-13 dashboard aggregation; US-40 baseline snapshots
Bansil	US-51 composer wiring for lint and pre-flight
Dhavni didi	Load test with Affan: messages, live analysis, checks, live-map freshness and 500-node map
Pranshu	US-13 dashboard UI; US-40 baselines UI; US-57 decision-rules settings
Abhijeet	US-45 suggestion panel (accept, edit, reject with reason, defer); US-38 inbox; US-06 settings UI
Abhyudit	Checkpoint 2 recall report (caught by AI, by rules, missed — including reworded conflicts); suggestion correctness study (30 rated)
Affan	Load test; Checkpoint 2 E2E of the full loop

Checkpoint 2 demo: post → live map → conflict issue → AI suggestions → owners pick one → change request with findings delta → decision owner approves → merge → snapshot comparison; plus the recall report.

9. Sprint 6 — Written Changes, Updated SRS & Governance (Weeks 12–13)

Stories: US-61 (LLM-generated updated SRS with review), US-41 (SRS export), US-59 (Open-items report), US-55 (Written change log), US-50 (Map preview & time slider), US-18 (Compliance protection — conditional), US-08 (Severity), US-15 (Feedback view), US-42 (Timeline), US-53 (Discussion, votes, watch) — 37 points. US-48 (Work items) is deferred. Feature freeze at the end of Week 13.

Name	Task
Dhavni didi	US-55 change log and summaries (primary owner); US-50 map preview and time slider; US-42 timeline
Bansil	US-61 updated SRS: readiness checklist, requirements inserted verbatim, section review, fidelity check, publish (primary owner); US-41 DOCX, PDF and Markdown export; US-59 open-items report
Parth	US-18 compliance protection queue and audit export (if in scope)
Keval	US-61 SRS section prompts; US-08 severity; US-15 feedback view; US-55 AI summary paragraph
Heet	Recall tuning for the rule checks (US-58)
Madhav	US-53 discussion, votes, watch and links; E2E support with Affan
Pranshu	US-61 Updated SRS screen; US-50, US-42 and US-55 UI
Abhijeet	US-18 compliance screens; US-15 feedback view; US-08 severity UI
Abhyudit	Full backend regression; final accuracy and recall run; SRS fidelity tests
Affan	Security tests (isolation, join codes, rate limits, injection in posts); accessibility of core flows; production smoke tests

10. Sprint 7 — Release (Week 14)

No new stories — fixes, demo preparation and rehearsal only.

Name	Task
Everyone	Bug bash on the deployed system (led by Affan); blocking fixes only
Parth & Dhavni didi	Fix blocking bugs; deployment freeze at the end of day 3; stability checks on backend and frontend
Abhyudit & Bansil	Demo data: one workspace, three projects with confirmed context, ten members in different roles, planted scenarios, a published SRS v1.0
Pranshu & Abhijeet	Demo script and two full rehearsals on the deployed system
Madhav, Heet & Keval	Final report: accuracy and recall, load results, usability findings, retrospective, future extensions
Everyone	Final demo

11. Load Check & Risk Note

Heaviest sprints: Sprint 2 and Sprint 5 (38 points each). In Sprint 2, Heet and Keval both own core parts of the 13-point US-05 while Dhavni didi builds real-time messaging. If Sprint 2 is behind by day 5, move US-07, US-49 and US-22 into Sprint 3 — US-05 without US-07 still produces usable results. Sprint 5 ends at Checkpoint 2 and carries the trust work (US-45, US-57, US-58); if it slips, US-50 and US-53 are dropped from Sprint 6 so these finish first.

Never cut: the document path (US-60, 01, 02, 05, 09, 61), the collaboration path (US-24, 25, 27, 29, 30, 31, 14, 33, 34, 35, 52, 45) and the trust layer (US-56, 57, 58). Should-priority items are cut first, in the order given in the epics document.

Frontend and testing capacity. Two frontend developers carry many screens, so a shared component kit is built in Sprint 0, and Bansil and Madhav wire some UI in Sprints 5–6. Testing (Abhyudit, Affan) is two people for ten developers: every developer writes unit tests for their own code, and Madhav joins E2E testing in Sprint 6. If a checkpoint slips, borrow a backend or frontend member for testing in that sprint only, rather than letting QA fall behind.

Velocity. Sprints are planned at 37–38 points, assuming 6–8 hours per member per week. Recalibrate after Sprint 1 with actual velocity and apply the cut order early if needed. No scope is added mid-sprint.